# Hartshorne Chapter III: Cohomology and Derived Functors (Expanded notes for the first lectures)

## 1 Orientation and motivation

These notes begin Chapter III of Hartshorne, with two goals:

(1) motivate *sheaf cohomology* (especially cohomology of quasi-coherent sheaves on schemes);

(2) develop the abstract machine of *derived functors* in an abelian category, emphasizing universal $\delta$-functors, effaceability, and injective resolutions.

The starting point is the global sections functor

$$\Gamma(X, -) : \{\text{sheaves on } X\} \to \mathbf{Ab}, \qquad \mathcal{F} \mapsto \mathcal{F}(X),$$

which is always *left exact* but rarely exact. The derived functors of $\Gamma$ will be denoted

$$H^i(X, \mathcal{F}) := R^i\Gamma(X, \mathcal{F}),$$

and these are the cohomology groups of $\mathcal{F}$.

### 1.1 Motivation: Maps to projective space

Let $X$ be a scheme (typically projective over a field $k$) and let $L$ be an invertible sheaf on $X$. A choice of a finite-dimensional subspace $V \subseteq H^0(X, L)$ gives the *evaluation map*

$$\mathrm{ev}_V : V \otimes_k \mathcal{O}_X \longrightarrow L, \qquad s \otimes f \mapsto f \cdot s, \tag{1}$$

and hence (where defined) a morphism $\varphi_V : X \dashrightarrow \mathbf{P}(V^\vee)$.

What makes cohomology indispensable is that essentially every geometric property of $\varphi_V$ can be reformulated as a *surjectivity problem for restriction maps of global sections*, and these surjectivity problems are controlled by $H^1$ of suitable ideal sheaves via the long exact sequence in cohomology.

#### 1.1.1 Restriction to a finite subscheme and the $H^1$-obstruction

Let $Z \hookrightarrow X$ be a closed subscheme with ideal sheaf $\mathcal{I}_Z \subseteq \mathcal{O}_X$. Tensoring

$$0 \to \mathcal{I}_Z \to \mathcal{O}_X \to \mathcal{O}_Z \to 0$$

by $L$ gives a short exact sequence

$$0 \longrightarrow \mathcal{I}_Z \otimes L \longrightarrow L \longrightarrow L|_Z \longrightarrow 0. \tag{2}$$

Applying $\Gamma(X, -)$ yields the beginning of a long exact sequence

$$0 \to H^0(X, \mathcal{I}_Z \otimes L) \to H^0(X, L) \xrightarrow{\mathrm{res}_Z} H^0(Z, L|_Z) \xrightarrow{\delta_Z} H^1(X, \mathcal{I}_Z \otimes L) \to \cdots . \tag{3}$$

Thus the *failure* of the restriction map $\mathrm{res}_Z$ to be surjective is measured by the connecting homomorphism $\delta_Z$; in particular:

**Lemma 1.1** (A cohomological surjectivity criterion)**.** *If $H^1(X, \mathcal{I}_Z \otimes L) = 0$, then $\mathrm{res}_Z : H^0(X, L) \to H^0(Z, L|_Z)$ is surjective.*

**Remark 1.2.** This lemma is the basic template: to show a linear system has good geometric properties, it is often enough to prove vanishing of $H^1(X, \mathcal{I}_Z \otimes L)$ for a suitable class of (small) subschemes $Z$. In projective geometry one then invokes vanishing theorems (e.g. Serre vanishing) to guarantee these $H^1$ groups vanish after twisting by a sufficiently positive line bundle.

### 1.1.2 Global generation (base-point-freeness) via $H^1(\mathcal{I}_x \otimes L)$

For a (closed) point $x \in X$, let $\mathcal{I}_x$ be its ideal sheaf. The restriction map in (3) for $Z = \{x\}$ is exactly the evaluation-at-$x$ map

$$H^0(X, L) \longrightarrow H^0(\{x\}, L|_x) \cong L \otimes k(x).$$

Therefore $L$ is globally generated at $x$ iff this map is surjective, and by Lemma 1.1 a sufficient condition is:

$$H^1(X, \mathcal{I}_x \otimes L) = 0 \quad \Rightarrow \quad L \text{ is generated by global sections at } x.$$

In practice, if $X$ is projective and $A$ is ample, Serre vanishing gives $H^1(X, \mathcal{I}_x \otimes L \otimes A^{\otimes n}) = 0$ for $n \gg 0$, uniformly in $x$ once one makes the dependence precise; hence sufficiently positive twists of $L$ become globally generated.

### 1.1.3 Separating points via $H^1(\mathcal{I}_{\{x,y\}} \otimes L)$

To separate two (distinct) points $x \neq y$, one wants surjectivity of

$$H^0(X, L) \longrightarrow H^0(\{x, y\}, L|_{\{x,y\}}) \cong (L \otimes k(x)) \oplus (L \otimes k(y)).$$

If this map is surjective, we can prescribe values at $x$ and $y$ independently, so in particular there exists a section vanishing at $x$ but not at $y$.

Let $Z = \{x, y\}$ (scheme-theoretic union). Then Lemma 1.1 gives:

$$H^1(X, \mathcal{I}_{\{x,y\}} \otimes L) = 0 \quad \Rightarrow \quad H^0(X, L) \text{ separates } x \text{ and } y.$$

### 1.1.4 Separating tangent vectors via $H^1(\mathcal{I}_{x^{(1)}} \otimes L)$

To control the *differential* of $\varphi_V$, one must separate first-order infinitesimal data. Let $x^{(1)} \subseteq X$ denote the first infinitesimal neighborhood of $x$,

$$x^{(1)} := \mathrm{Spec}(\mathcal{O}_{X,x}/\mathfrak{m}_x^2),$$

with ideal sheaf $\mathcal{I}_{x^{(1)}}$.

Surjectivity of the restriction map

$$H^0(X, L) \longrightarrow H^0(x^{(1)}, L|_{x^{(1)}})$$

means global sections can prescribe both the value and first-order behavior at $x$ (equivalently: $H^0(X, L)$ separates tangent vectors at $x$). Again,

$$H^1(X, \mathcal{I}_{x^{(1)}} \otimes L) = 0 \quad \Rightarrow \quad H^0(X, L) \text{ separates tangent directions at } x.$$

Consequently, once $L$ is base-point-free, cohomological control of the groups $H^1(X, \mathcal{I}_{x^{(1)}} \otimes L)$ provides a systematic way to prove that $\varphi_{|L|}$ is an immersion (and, on a proper scheme, often a closed immersion after checking injectivity/separation).

**Remark 1.3** (Very ampleness as a length-2 condition)**.** A useful reformulation is: $L$ is very ample if and only if, for every length-2 zero-dimensional subscheme $Z \subset X$ (including nonreduced ones such as $x^{(1)}$), the map

$$H^0(X, L) \to H^0(Z, L|_Z)$$

is surjective. The point is that length-2 subschemes simultaneously encode "two distinct points" and "a tangent direction at one point." Cohomology enters through (3): vanishing of $H^1(X, \mathcal{I}_Z \otimes L)$ for all such $Z$ forces very ampleness.

### 1.1.5 Equations of the image and cohomology of the ideal sheaf

Suppose $L$ is very ample and gives an embedding

$$i : X \hookrightarrow \mathbf{P}^N = \mathbf{P}(H^0(X, L)^\vee), \qquad \text{with } L \simeq i^*\mathcal{O}_{\mathbf{P}^N}(1).$$

Let $\mathcal{I}_X \subseteq \mathcal{O}_{\mathbf{P}^N}$ be the ideal sheaf of the image. Twisting the defining short exact sequence $0 \to \mathcal{I}_X \to \mathcal{O}_{\mathbf{P}^N} \to i_*\mathcal{O}_X \to 0$ by $\mathcal{O}_{\mathbf{P}^N}(d)$ gives

$$0 \to \mathcal{I}_X(d) \to \mathcal{O}_{\mathbf{P}^N}(d) \to i_*L^{\otimes d} \to 0. \tag{4}$$

Taking global sections yields

$$0 \to H^0(\mathbf{P}^N, \mathcal{I}_X(d)) \to H^0(\mathbf{P}^N, \mathcal{O}(d)) \to H^0(X, L^{\otimes d}) \to H^1(\mathbf{P}^N, \mathcal{I}_X(d)) \to \cdots. \tag{5}$$

Here:

- $H^0(\mathbf{P}^N, \mathcal{I}_X(d))$ is precisely the space of degree-$d$ homogeneous equations vanishing on $X$.
- The obstruction to the surjectivity of the restriction map $H^0(\mathbf{P}^N, \mathcal{O}(d)) \to H^0(X, L^{\otimes d})$ is measured by $H^1(\mathbf{P}^N, \mathcal{I}_X(d))$.

Thus cohomology controls not only whether a linear system gives a morphism/immersion/embedding, but also how complicated the embedded image is (its equations in each degree). This is one of the main reasons Hartshorne Chapter III quickly pivots from "defining cohomology" to "proving vanishing and finite generation theorems": those theorems translate directly into geometric statements about maps to projective space.

## 1.2 Motivation II: numerical invariants

Cohomology produces numbers:

$$h^i(X, \mathcal{F}) := \dim_k H^i(X, \mathcal{F}) \quad \text{(over a field, when finite dimensional)},$$

and their alternating sum

$$\chi(X, \mathcal{F}) := \sum_{i \geq 0} (-1)^i h^i(X, \mathcal{F}),$$

the *Euler characteristic*.

For projective varieties, coherent cohomology groups are typically finite dimensional over $k$, and the Euler characteristic is often the most robust invariant.

**Example 1.4** (Curves: genus)**.** If $C$ is a smooth projective curve over $k$, then

$$g(C) = \dim_k H^1(C, \mathcal{O}_C).$$

This is the (arithmetic) genus; it is a fundamental invariant in the classification of curves.

**Example 1.5** (Surfaces: $q$ and $p_g$)**.** If $X$ is a smooth projective surface over $k$, one defines

$$q(X) := \dim_k H^1(X, \mathcal{O}_X) \quad \text{(irregularity)}, \qquad p_g(X) := \dim_k H^2(X, \mathcal{O}_X) \quad \text{(geometric genus)}.$$

These invariants play a central role in the Enriques–Kodaira classification.

**Example 1.6** (Plurigenera)**.** For a smooth projective variety $X$ with canonical bundle $\omega_X$, the numbers

$$P_m(X) := \dim_k H^0(X, \omega_X^{\otimes m})$$

are the *plurigenera*. They are among the key birational invariants used in higher-dimensional classification.

## 1.3 Motivation III: intersection theory via Euler characteristics

On a smooth projective surface $X$, one can package intersection numbers in terms of Euler characteristics (and later justify/compute them using Riemann–Roch).

Let $C, D \subset X$ be (Cartier) divisors, and write $\mathcal{O}_X(-C)$ for the associated line bundle.

**Definition 1.7** (Intersection number via Euler characteristic (surface case))**.** Define

$$C \cdot D \;:=\; \chi\big(\mathcal{O}_X(-C-D)\big) - \chi\big(\mathcal{O}_X(-C)\big) - \chi\big(\mathcal{O}_X(-D)\big) + \chi(\mathcal{O}_X).$$

**Remark 1.8.** This definition is engineered so that $C \cdot D$ depends only on numerical equivalence classes and behaves well in flat families: Euler characteristics of coherent sheaves are stable in proper flat families (a form of cohomology and base change). One later proves that Definition 1.7 recovers the usual geometric intersection pairing.

## 1.4 Motivation IV: comparison with familiar cohomology

Over $\mathbb{C}$, coherent cohomology is tied to singular cohomology by Hodge theory. For a smooth projective complex variety $X$,

$$H^n(X^{an}, \mathbb{C}) \cong \bigoplus_{p+q=n} H^q(X, \Omega_X^p),$$

where $X^{an}$ is the associated complex manifold and $\Omega_X^p$ are algebraic differential forms. This is a major reason coherent cohomology is indispensable: it mediates between algebraic geometry and topology/analysis.

## 2 Global sections are left exact but not exact

Let $X$ be a topological space. Write $\mathbf{Ab}(X)$ for the category of sheaves of abelian groups on $X$; if $(X, \mathcal{O}_X)$ is a ringed space, write $\mathbf{Mod}(\mathcal{O}_X)$ for sheaves of $\mathcal{O}_X$-modules.

**Proposition 2.1.** *The global sections functor*

$$\Gamma(X, -) : \mathbf{Ab}(X) \to \mathbf{Ab} \qquad (\textit{and similarly } \Gamma(X, -) : \mathbf{Mod}(\mathcal{O}_X) \to \mathbf{Mod}(\Gamma(X, \mathcal{O}_X)))$$

*is left exact.*

*Proof.* Let $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}''$ be an exact sequence of sheaves (exact on stalks). For any open $U \subseteq X$, applying "sections on $U$" gives an exact sequence

$$0 \to \mathcal{F}'(U) \to \mathcal{F}(U) \to \mathcal{F}''(U),$$

because kernels of morphisms of sheaves are computed sectionwise. Taking $U = X$ yields left exactness of $\Gamma(X, -)$. □

Surjectivity on global sections generally fails.

**Example 2.2** (A standard projective example)**.** On $\mathbf{P}^1_k$ there is an exact sequence of locally free sheaves

$$0 \longrightarrow \mathcal{O}_{\mathbf{P}^1}(-2) \longrightarrow \mathcal{O}_{\mathbf{P}^1}(-1)^{\oplus 2} \longrightarrow \mathcal{O}_{\mathbf{P}^1} \longrightarrow 0.$$

Taking global sections gives

$$0 \longrightarrow H^0(\mathbf{P}^1, \mathcal{O}(-2)) \longrightarrow H^0(\mathbf{P}^1, \mathcal{O}(-1))^{\oplus 2} \longrightarrow H^0(\mathbf{P}^1, \mathcal{O}) \longrightarrow \cdots$$

but $H^0(\mathbf{P}^1, \mathcal{O}(-1)) = 0$, while $H^0(\mathbf{P}^1, \mathcal{O}) \cong k$, so the map $H^0(\mathbf{P}^1, \mathcal{O}(-1))^{\oplus 2} \to H^0(\mathbf{P}^1, \mathcal{O})$ is the zero map and is not surjective. Thus $\Gamma(\mathbf{P}^1, -)$ is not exact.

The remedy is to measure the failure of exactness by a sequence of functors $H^i(X, -)$, beginning with

$$H^0(X, -) = \Gamma(X, -).$$

## 3 Abelian categories and left exact functors

Derived functors live naturally in abelian categories.

**Definition 3.1** (Abelian category)**.** A category $\mathcal{A}$ is *abelian* if:

(1) $\mathrm{Hom}_{\mathcal{A}}(A, B)$ is an abelian group for all $A, B$, and composition is bilinear;

(2) $\mathcal{A}$ has a zero object, finite products and coproducts (which then coincide as *biproducts*);

(3) every morphism has a kernel and cokernel;

(4) for every morphism $f$, the canonical map $\mathrm{coim}(f) \to \mathrm{im}(f)$ is an isomorphism.

**Example 3.2.** (a) $\mathbf{Mod}(R)$ for a ring $R$ (modules over $R$).

(b) $\mathbf{Ab}(X)$ for a topological space $X$.

(c) $\mathbf{Mod}(\mathcal{O}_X)$ for a ringed space $(X, \mathcal{O}_X)$; in particular $\mathbf{QCoh}(X)$ and $\mathbf{Coh}(X)$ for a scheme $X$.

**Example 3.3** (Non-examples)**.** (a) Free abelian groups: the cokernel of $2\mathbf{Z} \hookrightarrow \mathbf{Z}$ is $\mathbf{Z}/2\mathbf{Z}$, not free.

(b) Hausdorff topological abelian groups: cokernels in the category of topological groups need not be Hausdorff; the abelian-category axioms fail in a serious way.

**Definition 3.4** (Left exact functor)**.** An additive functor $F : \mathcal{A} \to \mathcal{B}$ between abelian categories is *left exact* if it preserves finite limits, equivalently if for every short exact sequence

$$0 \to A' \to A \to A'' \to 0$$

the sequence

$$0 \to F(A') \to F(A) \to F(A'')$$

is exact.

**Remark 3.5.** Many geometric functors are left exact:

$\Gamma(X, -)$, $\quad f_*$ (pushforward of sheaves), $\quad \mathrm{Hom}_R(M, -)$ (in the second variable), $\quad (-)^G$ (group invariants)

Right exact functors (e.g. $- \otimes_R M$) lead instead to *left derived functors* and projective/flat resolutions (à la Tor).

# 4 $\delta$-functors and universality

## 4.1 Cohomological $\delta$-functors

**Definition 4.1** (Cohomological $\delta$-functor)**.** Let $\mathcal{A}, \mathcal{B}$ be abelian categories. A *(cohomological) $\delta$-functor* from $\mathcal{A}$ to $\mathcal{B}$ is a sequence of additive functors

$$T^i : \mathcal{A} \to \mathcal{B} \qquad (i \geq 0)$$

together with, for every short exact sequence $0 \to A' \to A \to A'' \to 0$ in $\mathcal{A}$, connecting morphisms

$$\delta^i : \ T^i(A'') \longrightarrow T^{i+1}(A')$$

such that:

(1) **Long exactness:** for each short exact sequence, the induced sequence

$$0 \to T^0(A') \to T^0(A) \to T^0(A'') \xrightarrow{\delta^0} T^1(A') \to T^1(A) \to T^1(A'') \xrightarrow{\delta^1} \cdots$$

is exact.

(2) **Naturality:** the connecting maps are functorial with respect to morphisms of short exact sequences.

**Remark 4.2.** Sheaf cohomology $H^i(X, -)$ is the prototype: given $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$, there is a natural long exact sequence in cohomology. The point of derived functors is that *this exactness is forced* from the general formalism.

### 4.2 Morphisms and universal $\delta$-functors

**Definition 4.3** (Morphism of $\delta$-functors)**.** A morphism $\varphi : T^* \to S^*$ of $\delta$-functors is a collection of natural transformations $\varphi^i : T^i \to S^i$ such that for every short exact sequence the squares involving $\delta^i$ commute, i.e. $\varphi^{i+1} \circ \delta^i_T = \delta^i_S \circ \varphi^i$.

**Definition 4.4** (Universal $\delta$-functor)**.** A $\delta$-functor $T^*$ is *universal* if for every $\delta$-functor $S^*$ and every natural transformation $\psi^0 : T^0 \to S^0$, there exists a unique morphism of $\delta$-functors $\psi^* : T^* \to S^*$ extending $\psi^0$.

**Remark 4.5.** Universality is the categorical formulation of: *"there is only one reasonable way to add higher functors to a given left exact functor so as to obtain long exact sequences."*

## 5 Effaceability and dimension shifting

### 5.1 Effaceability

**Definition 5.1** (Effaceable)**.** A $\delta$-functor $T^*$ is *effaceable* if for every object $A \in \mathcal{A}$ and every $i > 0$, there exists a monomorphism $f : A \hookrightarrow B$ in $\mathcal{A}$ such that

$$T^i(f) : T^i(A) \to T^i(B) \quad \text{is the zero map.}$$

Effaceability is the key hypothesis guaranteeing universality.

**Theorem 5.2** (Effaceable $\Rightarrow$ universal)**.** *Every effaceable cohomological $\delta$-functor is universal.*

**Remark 5.3.** Hartshorne states this as a basic formal result; the proof is a standard (and instructive) "dimension shifting + diagram chase" argument. A good first exposure is to do the proof at least once carefully.

### 5.2 Computing by induction if one can efface

Suppose $T^*$ is a cohomological $\delta$-functor and we can efface $T^i$ for $i > 0$ by a monomorphism $A \hookrightarrow B$. Let $C := B/A$ (so $0 \to A \to B \to C \to 0$ is exact). Then the long exact sequence gives:

- Since $T^i(A) \to T^i(B)$ is zero (for $i > 0$), exactness shows that

$$T^i(A) \cong \operatorname{coker}\big(T^{i-1}(B) \to T^{i-1}(C)\big) \qquad (i \geq 1).$$

- In particular,

$$T^1(A) = \operatorname{coker}\big(T^0(B) \to T^0(C)\big),$$

  and

$$T^2(A) = \operatorname{coker}\big(T^1(B) \to T^1(C)\big),$$

  etc.

This is the "inductive computation" viewpoint: if we can keep embedding into objects with vanishing higher cohomology, we can compute everything from degree 0 data.

# 6 Injective objects and enough injectives

## 6.1 Injectives

**Definition 6.1** (Injective object)**.** An object $I \in \mathcal{A}$ is *injective* if for every monomorphism $u : X \hookrightarrow Y$ and every morphism $f : X \to I$, there exists $\tilde{f} : Y \to I$ with $\tilde{f} \circ u = f$.

**Remark 6.2.** Equivalently, $I$ is injective if and only if the contravariant functor $\mathrm{Hom}_{\mathcal{A}}(-, I)$ is exact (it is always left exact).

**Definition 6.3** (Enough injectives)**.** An abelian category $\mathcal{A}$ has *enough injectives* if every object $A \in \mathcal{A}$ admits a monomorphism $A \hookrightarrow I$ with $I$ injective.

## 6.2 Effaceability forces vanishing on injectives

**Proposition 6.4.** *Let $T^*$ be an effaceable $\delta$-functor on an abelian category $\mathcal{A}$. If $I$ is injective in $\mathcal{A}$, then $T^i(I) = 0$ for all $i > 0$.*

*Proof.* Fix $i > 0$. By effaceability, choose a monomorphism $f : I \hookrightarrow B$ such that $T^i(f) = 0$. Let $C := B/I$, so $0 \to I \xrightarrow{f} B \to C \to 0$ is exact.

Since $I$ is injective, this short exact sequence splits; hence there exists a retraction $p : B \to I$ with $p \circ f = \mathrm{id}_I$.

Apply $T^i$:

$$T^i(p) \circ T^i(f) = T^i(p \circ f) = T^i(\mathrm{id}_I) = \mathrm{id}_{T^i(I)}.$$

But $T^i(f) = 0$, so the left-hand side is 0. Therefore $\mathrm{id}_{T^i(I)} = 0$, hence $T^i(I) = 0$. □

# 7 Derived functors via injective resolutions

We now explain the construction of right derived functors of a left exact functor $F$. This is the abstract framework underlying sheaf cohomology.

## 7.1 Injective resolutions

**Definition 7.1** (Injective resolution)**.** Let $\mathcal{A}$ be an abelian category with enough injectives. An *injective resolution* of $A \in \mathcal{A}$ is an exact complex

$$0 \longrightarrow A \longrightarrow I^0 \longrightarrow I^1 \longrightarrow I^2 \longrightarrow \cdots$$

with each $I^n$ injective.

Existence is proved inductively: embed $A$ into an injective $I^0$, set $C^0 := I^0/A$, embed $C^0$ into an injective $I^1$, etc.

## 7.2 Comparison theorem (functoriality up to homotopy)

The basic tool is:

**Theorem 7.2** (Comparison theorem)**.** *Let $A, B \in \mathcal{A}$ and choose injective resolutions $A \to I^\bullet$, $B \to J^\bullet$. Any morphism $f : A \to B$ extends to a chain map $I^\bullet \to J^\bullet$, and any two such extensions are chain homotopic.*

**Remark 7.3.** This result is often proved by induction using injectivity to extend maps step-by-step, and then a second induction to produce the homotopy between two extensions. The "unique up to homotopy" clause is exactly what is needed to ensure derived functors are well-defined.

## 7.3 Definition of right derived functors

**Definition 7.4** (Right derived functors)**.** Let $F : \mathcal{A} \to \mathcal{B}$ be a left exact functor, with $\mathcal{A}$ having enough injectives. For $A \in \mathcal{A}$ choose an injective resolution $A \to I^\bullet$ and set

$$R^i F(A) := H^i\big(F(I^\bullet)\big) \qquad (i \geq 0),$$

the cohomology of the cochain complex $F(I^\bullet)$.

**Proposition 7.5** (Well-definedness)**.** *$R^i F(A)$ is independent of the chosen injective resolution, up to canonical isomorphism. Moreover $R^i F$ is functorial in $A$.*

*Sketch.* Given two injective resolutions of $A$, the comparison theorem produces chain maps between them inducing mutually inverse isomorphisms on cohomology, and these isomorphisms do not depend on choices because of uniqueness up to homotopy. Functoriality in $A$ is obtained by extending maps $A \to B$ to maps of resolutions and passing to cohomology; uniqueness up to homotopy ensures the induced map on $H^i$ is independent of the extension. □

## 7.4 The basic properties

**Proposition 7.6.** *Let $F$ be left exact.*

*(1)* $R^0 F \simeq F$.

*(2) For every short exact sequence $0 \to A' \to A \to A'' \to 0$ in $\mathcal{A}$ there is a natural long exact sequence*

$$0 \to F(A') \to F(A) \to F(A'') \to R^1 F(A') \to R^1 F(A) \to R^1 F(A'') \to R^2 F(A') \to \cdots .$$

*(3)* $R^i F(I) = 0$ *for $i > 0$ and $I$ injective.*

*Idea.* (1) follows because the beginning of an injective resolution is left exact and $F$ is left exact.

(2) is proved by choosing injective resolutions and applying the *horseshoe lemma* to obtain a short exact sequence of complexes $0 \to I'^\bullet \to I^\bullet \to I''^\bullet \to 0$, then using the long exact sequence in cohomology of complexes.

(3) If $I$ is injective, use the resolution $0 \to I \to I \to 0 \to \cdots$, so $F(I^\bullet)$ has cohomology only in degree 0. □

Thus $R^* F$ forms a cohomological $\delta$-functor extending $F$.

## 7.5 Universality of derived functors

**Proposition 7.7.** *If $\mathcal{A}$ has enough injectives, then $R^* F$ is an effaceable $\delta$-functor, hence universal.*

*Proof.* Let $A \in \mathcal{A}$ and choose a monomorphism $A \hookrightarrow I$ with $I$ injective. Then for $i > 0$ the induced map

$$R^i F(A) \to R^i F(I)$$

is the zero map because $R^i F(I) = 0$. Hence $R^* F$ is effaceable. By Theorem 5.2, it is universal. □

**Remark 7.8.** This is the main conceptual payoff: *derived functors are the universal way to repair the failure of exactness of a left exact functor.*

# 8 Acyclic objects and acyclic resolutions

Injective resolutions always exist (in good categories), but they can be unwieldy. Often one computes derived functors using a nicer resolution by "acyclic" objects.

**Definition 8.1** ($F$-acyclic)**.** Let $F : \mathcal{A} \to \mathcal{B}$ be left exact with right derived functors $R^iF$. An object $A \in \mathcal{A}$ is *$F$-acyclic* if $R^iF(A) = 0$ for all $i > 0$.

**Definition 8.2** (Acyclic resolution)**.** An *$F$-acyclic resolution* of $A$ is an exact complex

$$0 \to A \to A^0 \to A^1 \to A^2 \to \cdots$$

with each $A^n$ $F$-acyclic.

**Theorem 8.3** (Computing with acyclic resolutions)**.** *If $0 \to A \to A^\bullet$ is an $F$-acyclic resolution, then*

$$R^iF(A) \cong H^i\big(F(A^\bullet)\big) \qquad \textit{for all } i \geq 0.$$

**Remark 8.4.** For sheaf cohomology, common acyclic objects include injective sheaves, flasque (flabby) sheaves, and (on nice spaces) soft or fine sheaves. Cech resolutions are another computational tool under separation hypotheses.

# 9 Examples of derived functors

## 9.1 Sheaf cohomology

Let $X$ be a topological space. The category $\mathbf{Ab}(X)$ has enough injectives (proved below), and we define:

$$H^i(X, \mathcal{F}) := R^i\Gamma(X, \mathcal{F}).$$

The long exact sequence of Proposition 7.6 is the usual long exact cohomology sequence.

Similarly, on a ringed space $(X, \mathcal{O}_X)$, the category $\mathbf{Mod}(\mathcal{O}_X)$ has enough injectives, and for an $\mathcal{O}_X$-module $\mathcal{F}$:

$$H^i(X, \mathcal{F}) := R^i\Gamma(X, \mathcal{F}) \quad \text{(cohomology of sheaves of modules)}.$$

**Remark 9.1** (A first answer to "when is $\Gamma$ exact?")**.** Given $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$, the induced sequence

$$0 \to \Gamma(X, \mathcal{F}') \to \Gamma(X, \mathcal{F}) \to \Gamma(X, \mathcal{F}'') \to 0$$

is exact if and only if the connecting homomorphism

$$\Gamma(X, \mathcal{F}'') \longrightarrow H^1(X, \mathcal{F}')$$

vanishes; in particular, it is exact whenever $H^1(X, \mathcal{F}') = 0$. More broadly, vanishing of higher cohomology is the condition under which $\Gamma$ behaves like an exact functor.

## 9.2 Derived pushforward

For a continuous map of ringed spaces (or schemes) $f : X \to Y$, the pushforward

$$f_* : \mathbf{Ab}(X) \to \mathbf{Ab}(Y) \qquad (\text{or } \mathbf{Mod}(\mathcal{O}_X) \to \mathbf{Mod}(\mathcal{O}_Y))$$

is left exact. Its right derived functors are denoted

$$R^if_*.$$

These package all higher direct images and are central in Leray spectral sequences, cohomology and base change, and many foundational results (e.g. proper mapping theorems).

### 9.3 Ext and group cohomology

**Example 9.2** (Ext)**.** In $\mathbf{Mod}(R)$, the functor $\mathrm{Hom}_R(M,-)$ is left exact. Its right derived functors are

$$R^i\mathrm{Hom}_R(M,-) = \mathrm{Ext}^i_R(M,-).$$

(Here we are deriving in the *second* variable; deriving in the first variable uses projective resolutions instead.)

**Example 9.3** (Group cohomology)**.** Let $G$ be a group and consider the category of $\mathbf{Z}[G]$-modules. The invariants functor $(-)^G$ is left exact. Its derived functors give group cohomology:

$$H^i(G,M) = R^i(M \mapsto M^G).$$

## 10 Enough injectives for sheaves of modules

We now justify a crucial input: categories of sheaves of modules have enough injectives.

**Proposition 10.1.** *Let $(X,\mathcal{O}_X)$ be a ringed space. The abelian category $\mathbf{Mod}(\mathcal{O}_X)$ has enough injectives.*

*Proof.* We give a standard construction.

**Step 1: embed stalkwise into injectives.** Fix $\mathcal{F} \in \mathbf{Mod}(\mathcal{O}_X)$. For each point $x \in X$, the stalk $\mathcal{F}_x$ is an $\mathcal{O}_{X,x}$-module. Using the fact that $\mathbf{Mod}(\mathcal{O}_{X,x})$ has enough injectives (a theorem in module theory), choose a monomorphism

$$\iota_x : \mathcal{F}_x \hookrightarrow I_x$$

with $I_x$ an injective $\mathcal{O}_{X,x}$-module.

**Step 2: build an injective sheaf from skyscrapers.** Let $j_x : \{x\} \hookrightarrow X$ be the inclusion. Define the skyscraper $\mathcal{O}_X$-module $j_{x*}I_x$ by

$$(j_{x*}I_x)(U) = \begin{cases} I_x & x \in U, \\ 0 & x \notin U. \end{cases}$$

This is naturally an $\mathcal{O}_X$-module, and any morphism $\mathcal{G} \to j_{x*}I_x$ is determined by its induced map on the stalk at $x$. Concretely,

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{G}, j_{x*}I_x) \;\cong\; \mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{G}_x, I_x). \tag{6}$$

Since $I_x$ is injective, the functor $\mathrm{Hom}_{\mathcal{O}_{X,x}}(-,I_x)$ is exact; by (6), the functor $\mathrm{Hom}_{\mathcal{O}_X}(-,j_{x*}I_x)$ is exact. Hence $j_{x*}I_x$ is injective in $\mathbf{Mod}(\mathcal{O}_X)$.

Now set

$$\mathcal{I} := \prod_{x\in X} j_{x*}I_x.$$

Products of injectives are injective: for any $\mathcal{G}$,

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{G},\mathcal{I}) \cong \prod_{x\in X} \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{G}, j_{x*}I_x),$$

and a product of surjections of abelian groups is surjective. Therefore $\mathcal{I}$ is injective.

**Step 3: embed $\mathcal{F}$ into $\mathcal{I}$.** For each $x$, the inclusion $\iota_x : \mathcal{F}_x \hookrightarrow I_x$ corresponds (via (6)) to a morphism of sheaves $\mathcal{F} \to j_{x*}I_x$. By the universal property of the product, these assemble into a morphism

$$\mathcal{F} \longrightarrow \prod_{x \in X} j_{x*}I_x = \mathcal{I}.$$

On stalks at $x$ this map is precisely $\mathcal{F}_x \hookrightarrow I_x$, hence injective. Since monomorphisms of sheaves are detected on stalks, $\mathcal{F} \hookrightarrow \mathcal{I}$ is a monomorphism into an injective object.

Thus $\mathbf{Mod}(\mathcal{O}_X)$ has enough injectives. □

**Corollary 10.2.** *For any topological space $X$, the category $\mathbf{Ab}(X)$ of sheaves of abelian groups has enough injectives.*

*Proof.* A sheaf of abelian groups is the same thing as a sheaf of $\mathbf{Z}$-modules, i.e. an $\mathcal{O}_X$-module for the constant sheaf of rings $\mathcal{O}_X = \mathbf{Z}_X$. Apply Proposition 10.1. □

## 11 Cohomology as the derived functors of global sections

We now formally define sheaf cohomology.

**Definition 11.1** (Sheaf cohomology)**.** Let $X$ be a topological space and $\mathcal{F} \in \mathbf{Ab}(X)$. Choose an injective resolution

$$0 \to \mathcal{F} \to \mathcal{I}^0 \to \mathcal{I}^1 \to \cdots$$

and define

$$H^i(X, \mathcal{F}) := H^i\big(\Gamma(X, \mathcal{I}^\bullet)\big).$$

If $(X, \mathcal{O}_X)$ is a ringed space and $\mathcal{F} \in \mathbf{Mod}(\mathcal{O}_X)$, define $H^i(X, \mathcal{F})$ similarly.

**Proposition 11.2.** *$H^*(X, -)$ is a cohomological $\delta$-functor with $H^0(X, -) = \Gamma(X, -)$. It vanishes on injective sheaves in degrees $> 0$, is effaceable, and is universal.*

*Proof.* This is Proposition 7.6 and Proposition 7.7 applied to $F = \Gamma(X, -)$. □

**Remark 11.3** (Preview: quasi-coherent cohomology)**.** When $X$ is a scheme, one is especially interested in $\mathcal{F} \in \mathbf{QCoh}(X)$ (or $\mathbf{Coh}(X)$ when $X$ is Noetherian). Many foundational results in Hartshorne III concern:

- vanishing on affines ($H^i(X, \mathcal{F}) = 0$ for $i > 0$ and $\mathcal{F}$ quasi-coherent on $X$ affine);
- Cech computations for separated schemes;
- finiteness for projective schemes;
- Serre vanishing and Serre duality.

These will be developed after the abstract formalism is in place.

## 12 Suggested exercises for the first lectures

**Exercise 12.1** (Effaceable implies universal)**.** Prove Theorem 5.2: given an effaceable $\delta$-functor $T^*$ and another $\delta$-functor $S^*$, show any natural map $T^0 \to S^0$ extends uniquely to a morphism $T^* \to S^*$. (Hint: use dimension shifting from an effacing monomorphism $A \hookrightarrow B$ and the long exact sequences.)

**Exercise 12.2** (Comparison theorem)**.** Prove the comparison theorem (Theorem 7.2) by inductively extending a map $A \to B$ to maps $I^n \to J^n$ using injectivity, and then proving uniqueness up to homotopy.